



IIT Guwahati

Monday, 20 April 2026

CL625

MEMS Inspired Capacitive Pressure Sensor

GROUP 6

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MEMS - Cantilever Beams

Cantilever Sensor

01

Pressure causes a micro-cantilever beam to bend, producing changes in resistance and capacitance.

Fabrication

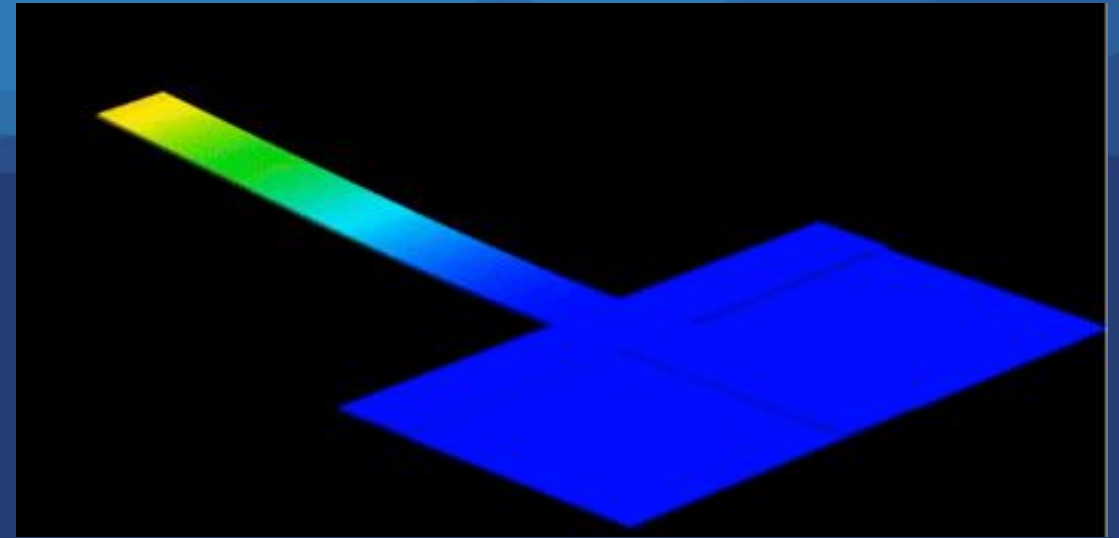
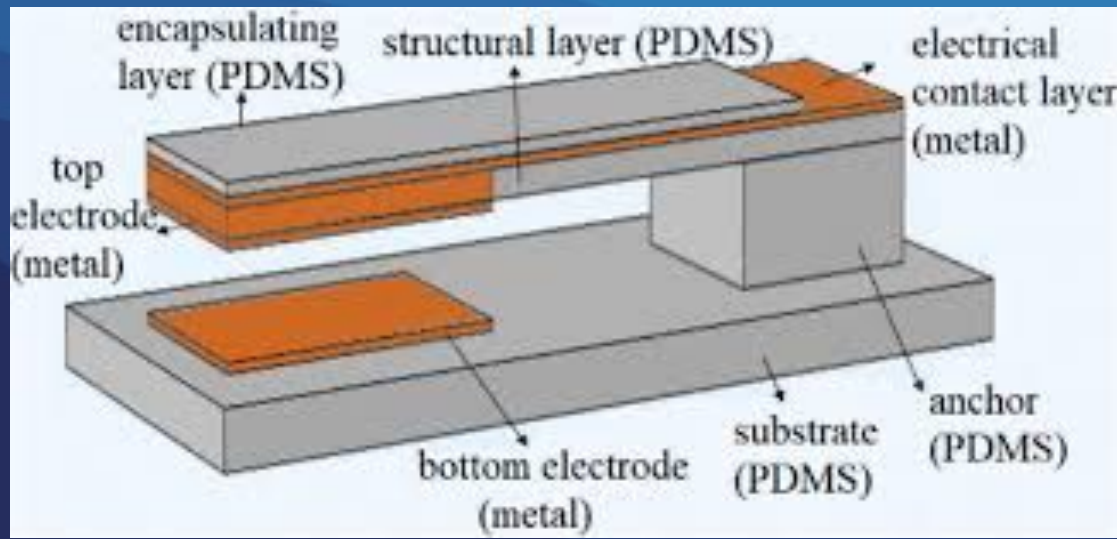
02

Micro-Nano Electro-Mechanical Systems are miniaturized devices fabricated on silicon chips.

Development

03

In recent developments, microfabrication and lithography enable highly miniaturized sensors.



Proof of Concept

Concept

Capacitance relation:

$$C_0 = \epsilon_0 \epsilon_r A / g_0$$

Cantilever tip
deflection under
force:

$$\delta = FL^3 / 3EI$$

Capacitance change due to deflection

01

New capacitance becomes

$$C = \epsilon_0 \epsilon_r A / (g_0 - \delta)$$

02

Change in capacitance:

$$\Delta C = \epsilon_0 \epsilon_r A / (g_0 - \delta) - \epsilon_0 \epsilon_r A / g_0$$

03

Using small-deflection
approximation:

$$\Delta C \approx \epsilon_0 \epsilon_r A \delta / g_0^2$$

Theory

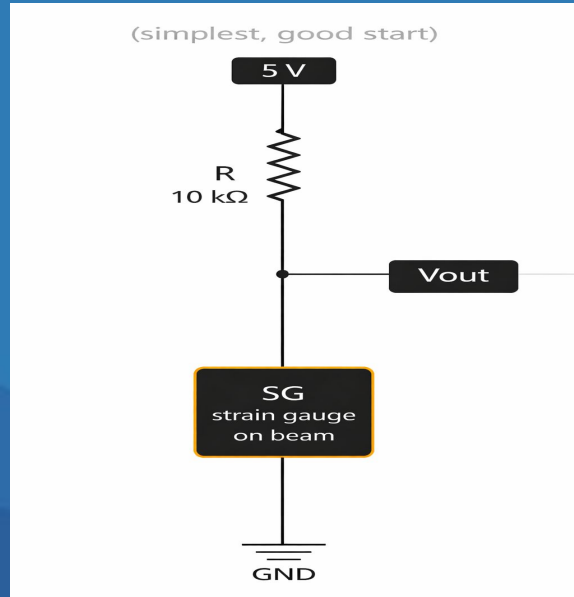
Using the **tapered capacitance** model the electrostatic force will be:

$$F_e = (1/2)V^2 (\epsilon w / \alpha) [1/(g_0 - x) - 1/(g_0 + \alpha L - x)]$$

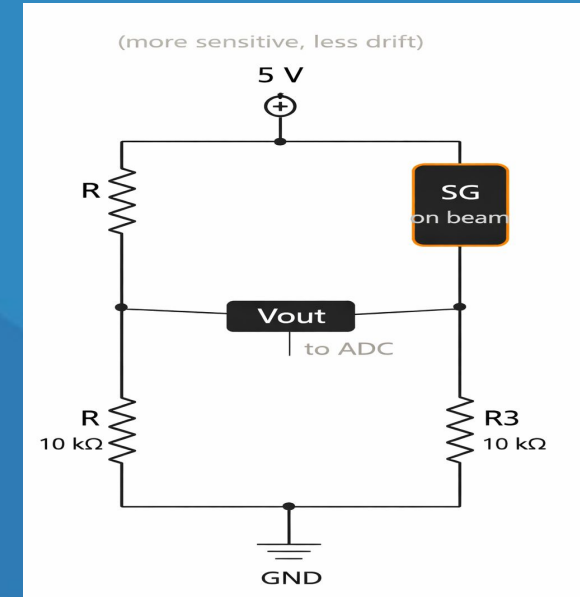
The result is higher sensitivity and improved stability.

Device proposition - Circuit diagram

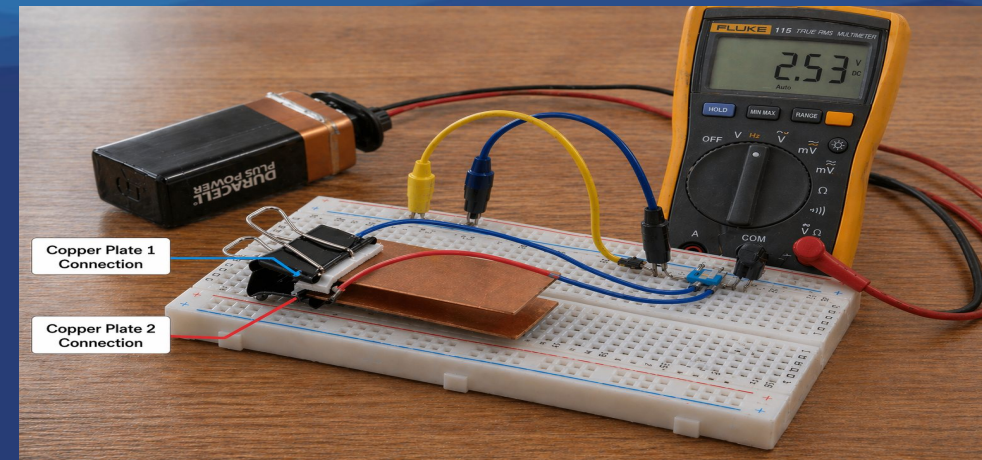
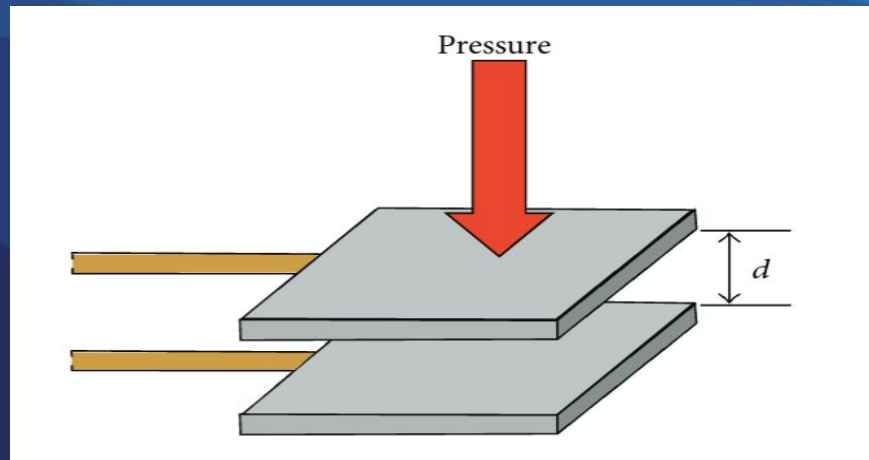
Voltage divider
(digital)



To ADC
(Analog-Digital
Converter)



Wheatstone
bridge (analog)



Multiple design fallacy

Option A — voltage divider

PROS

- + Minimal components—1 resistor + strain gauge
- + Easiest to wire and breadboard
- + Direct voltage read on any ADC pin
- + Ideal for quick prototyping and demos

CONS

- Output drifts with temperature changes
- Lower sensitivity — small flex changes hard to detect
- More noise in the signal
- Not suitable for precise or calibrated measurements

Option B — Wheatstone bridge

Recommended

PROS

- + Much higher sensitivity—detects tiny deflections
- + Self-compensating for temperature drift
- + Balanced output — starts at 0 V (easy to amplify)
- + Industry-standard for load cell and sensor design

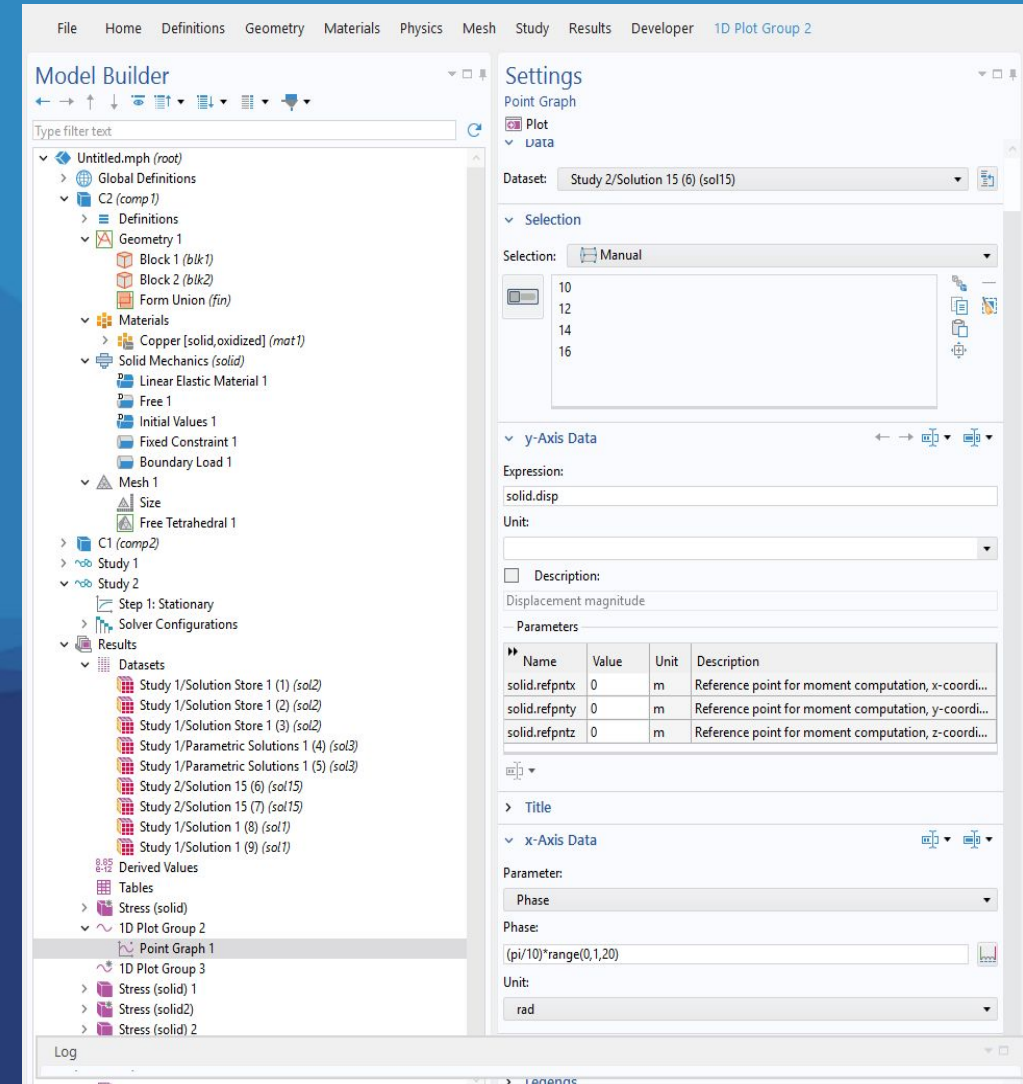
CONS

- Requires 3 matched resistors (same value)
- Slightly more complex to wire
- May need an op-amp to amplify small differential output
- Slightly higher component cost

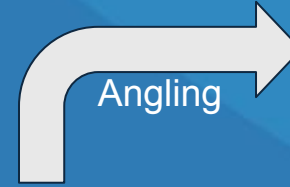
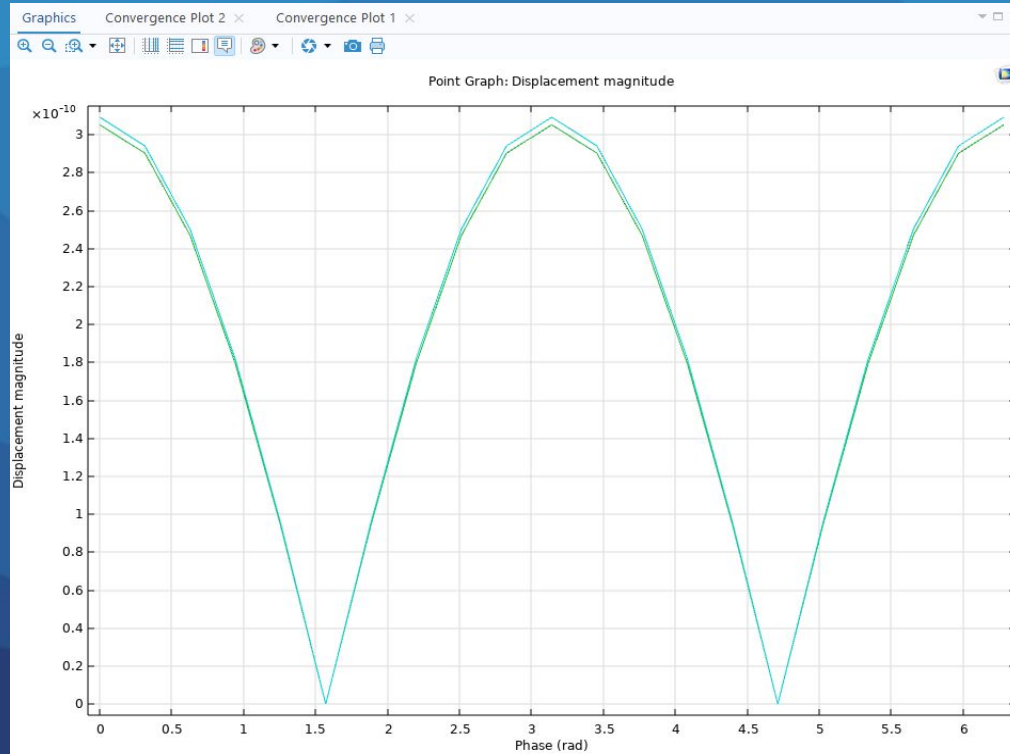
Simulation - COMSOL

COMSOL Multiphysics for MEMS Simulation

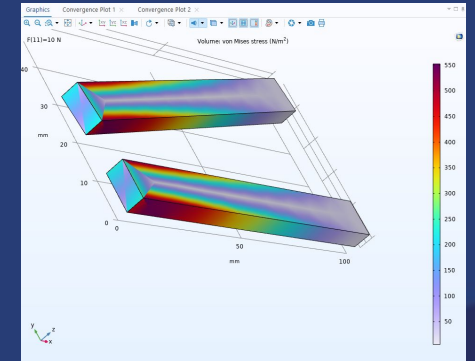
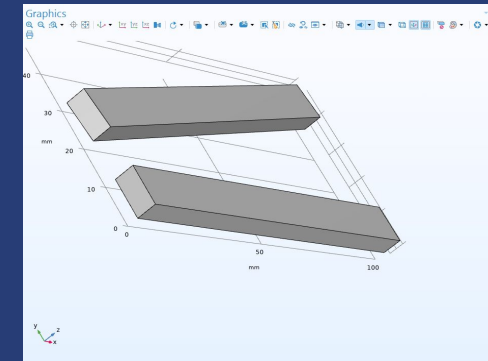
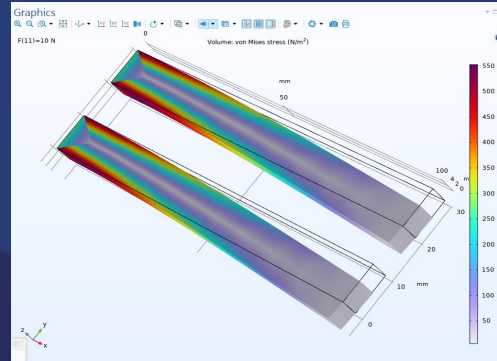
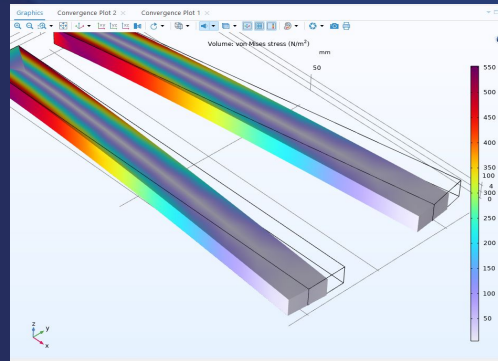
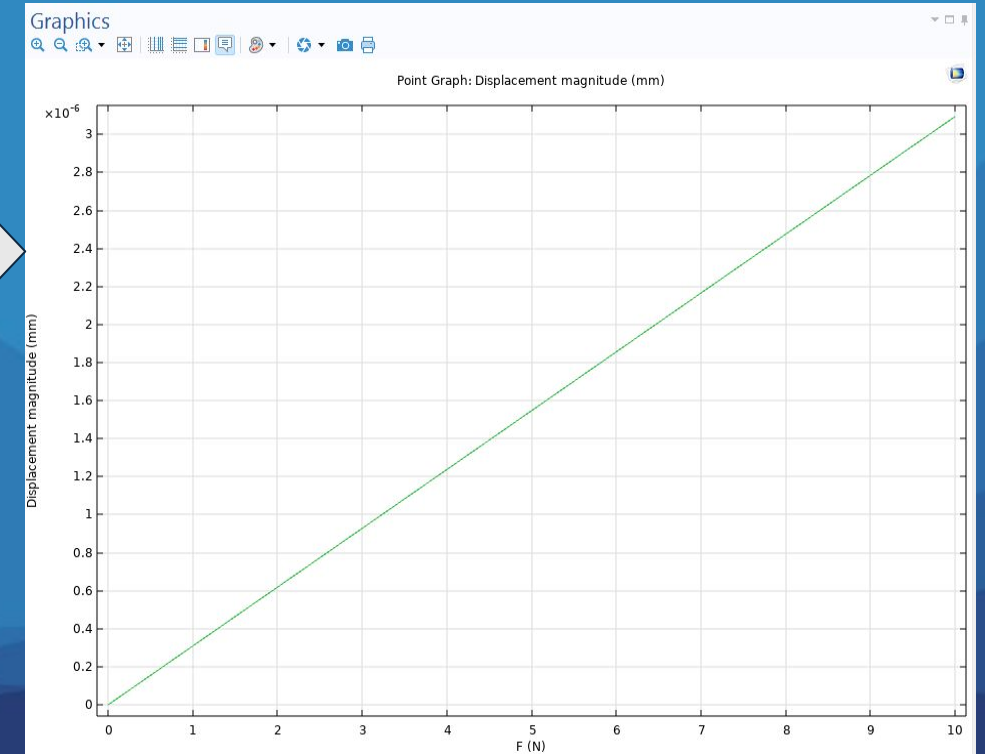
- **What is COMSOL?**
Finite element analysis (FEA) software is used to simulate mechanical, electrical, thermal, and fluidic behavior in a single platform.
- **Why COMSOL for MEMS?**
MEMS devices involve strong multiphysics coupling. COMSOL handles nonlinear effects like large deflection and fringing fields accurately.
- **Why did we use it?**
To simulate cantilever beam deformation under applied forces, analyze linearity improvement with beam angle, and study stress distribution, displacement, and sensitivity.
- **Key Advantage**
Enables realistic and high-accuracy prediction of MEMS behavior before fabrication, reducing time and cost.



Simulation results



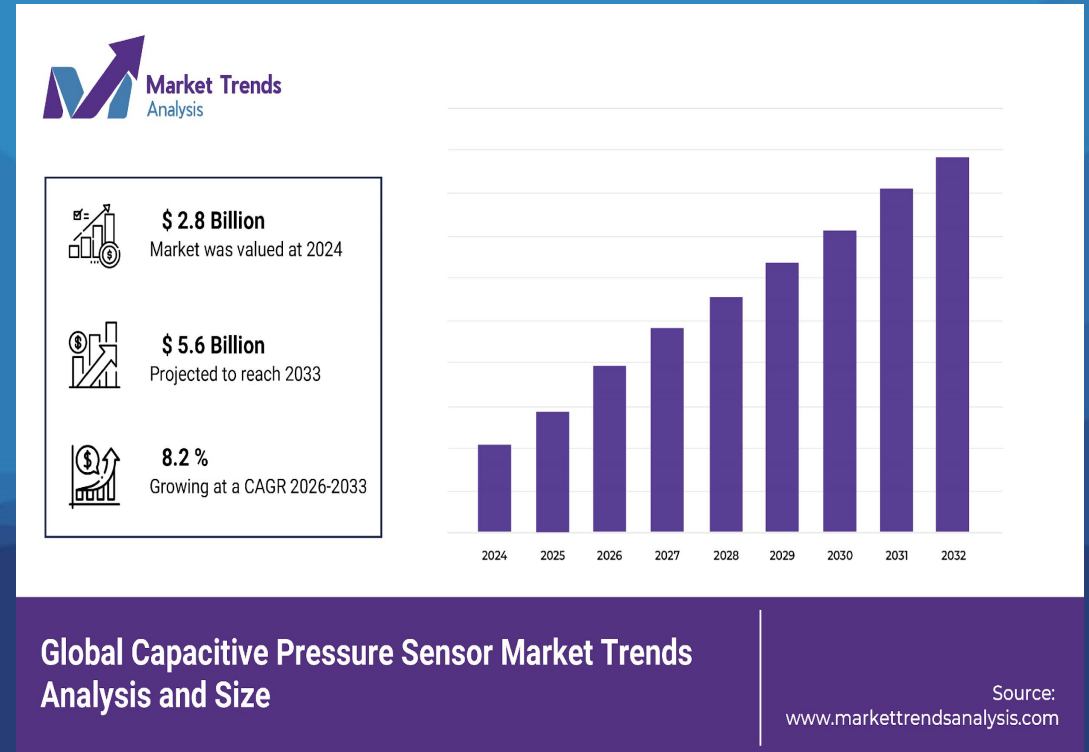
Displ. vs
Phase
Curves



Product and Marketing Strategy

- **The Problem:** Pull-in instability, limiting their usable linear sensing range to just ~33% of the initial gap.
- **Our Proprietary Solution:** Utilize a geometry-based non-parallel (tapered) electrode design and delay pull-in instability.
- **Target Market & Beachhead Strategy:** *Primary Focus:* Industrial IoT (IIoT) and smart manufacturing.
 - **Pilot Testing:** Deploying our MVP (Minimum Viable Product) in setups, such as machining units, will provide immediate data to validate our tech.
 - **Expansion:** Scaling into aerospace instrumentation and biomedical monitoring.

Value for Customers: Reduces equipment downtime, saving factories millions in operational costs.



Execution Roadmap

Revenue Streams (Monetization):

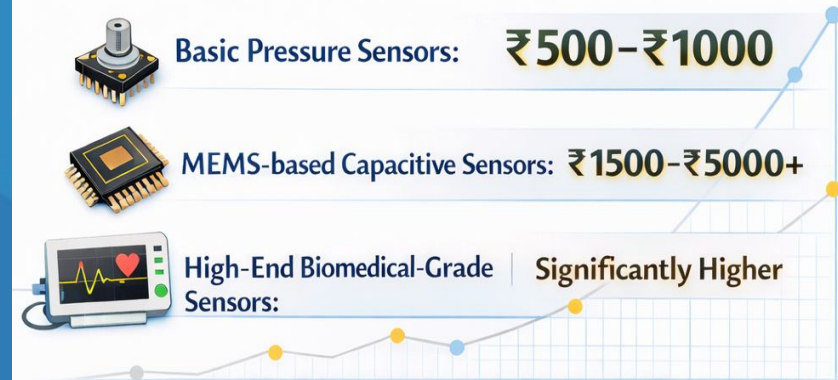
- **Hardware Sales:** Direct B2B sales of high-sensitivity sensor
- **Data & Analytics (SaaS):** Real-time pressure data analysis, enabled by IoT integration.

Operations & Fabrication Strategy:

- **Phase 1 (Months 1–6):** Develop sensors using photolithography and surface micromachining.
- **Phase 2 (Months 6–18):** Collaborate with MEMS foundries to scale production, ensuring miniaturization.

Funding Milestones: Apply for early academic grants and government startup programs (e.g., **Startup India**) to fund the MVP.

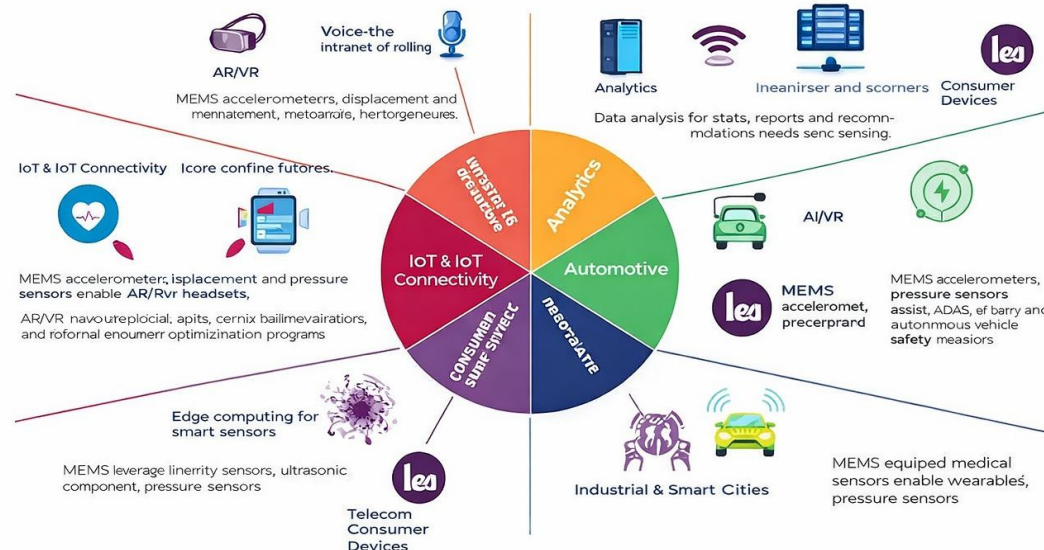
Comparable Market Pricing (Capacitive Pressure Sensors):



Our Pricing Strategy



✓ Affordable, competitive, and offers higher sensitivity compared to existing solutions



THANK YOU!

WHEN THEY SAY THEY HAD A BAD EXPERIENCE



WITH YOUR COMPETITOR...